

ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

Q.1 Create a table listing 5 key cybersecurity terms (such as threat, vulnerability, risk, asset, attack vector) and write a one-line definition for each in your own words.

ANS :

Cybersecurity Term	One-line Definition
Threat	Anything that could potentially cause harm to a system, network, or data.
Vulnerability	A weakness in a system that an attacker could exploit.
Risk	The possibility of harm or loss when a threat takes advantage of a vulnerability.
Asset	Anything valuable that needs protection, such as data, devices, or software.
Attack Vector	The method or path an attacker uses to gain unauthorized access to a system.

ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

Q.2 Research and find a real-world news article from the last 2 years about a major cyberattack on an Indian company or app (for example, a breach involving Zomato, Paytm, or IRCTC). Summarize what happened and identify one threat, one vulnerability, and one risk mentioned in the incident.

ANS :

Recent Cyberattack: Renova Paytech

A Noida-based Indian fintech company, **Renova Paytech Pvt. Ltd.**, reported a cyberattack in September 2026. According to *India Today*, the company's server was hacked on **September 13, 2026**, after which attackers used its API portal to make **497 unauthorized transactions**, transferring approximately **₹2.42 crore** to 58 bank accounts across 28 banks. Police registered a case and are investigating how the attackers obtained access to the server and API. *India Today*

India Today — Noida fintech firm server hacked, ₹2.42 crore siphoned via API

Threat, Vulnerability and Risk

Cybersecurity Term	What it means in this incident
Threat	Cybercriminals who hacked Renova Paytech's server and used the company's payment API to carry out unauthorized transactions.
Vulnerability	Weakness in the company's server/API security that allowed attackers to gain access; investigators are still examining exactly how the server was compromised. <i>India Today</i>
Risk	Financial loss and fraudulent transfer of company funds , with about ₹2.42 crore moved through 497 transactions. <i>India Today</i>

Short Summary

What happened: Renova Paytech's server was compromised on September 13, 2026. Attackers gained access to its API portal and transferred **₹2.42 crore through 497 transactions** to multiple bank accounts. The cybercrime police are investigating the breach and tracing the money. *India Today*

Threat → Cybercriminals

Vulnerability → Compromised server/API access

Risk → ₹2.42 crore financial loss and fraudulent transactions

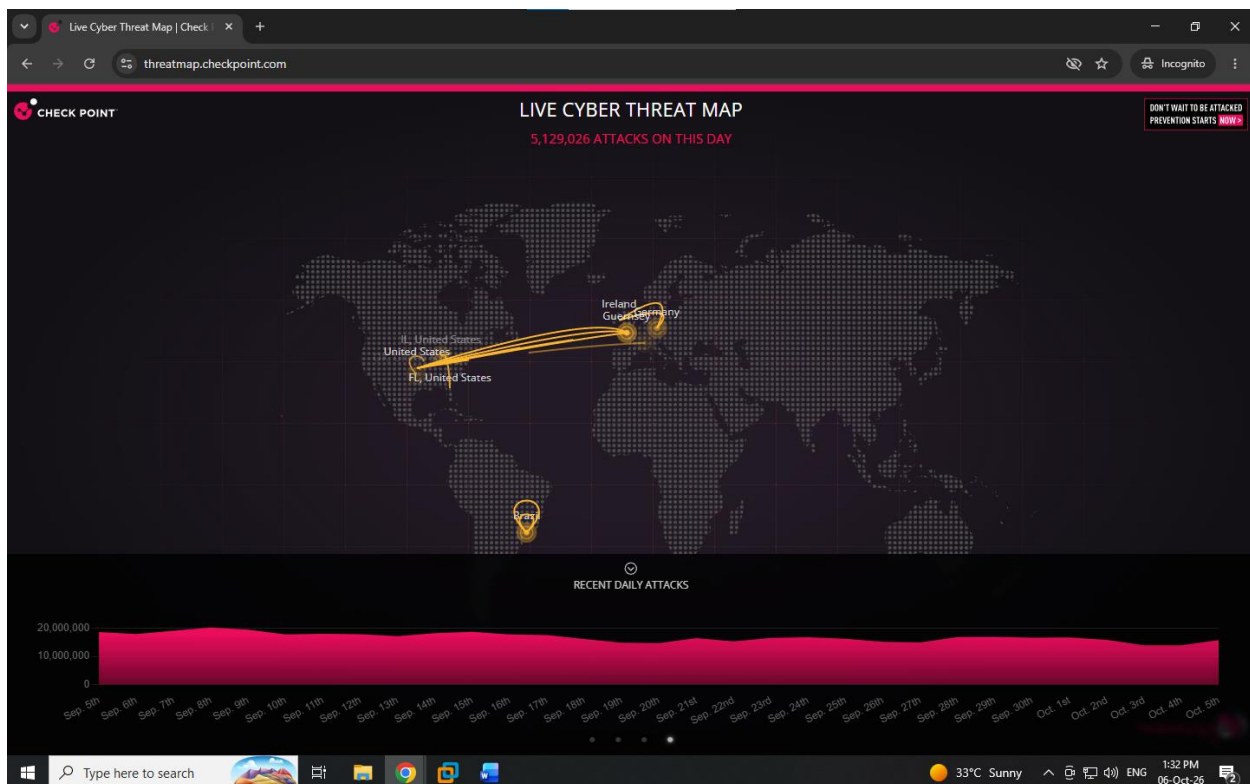
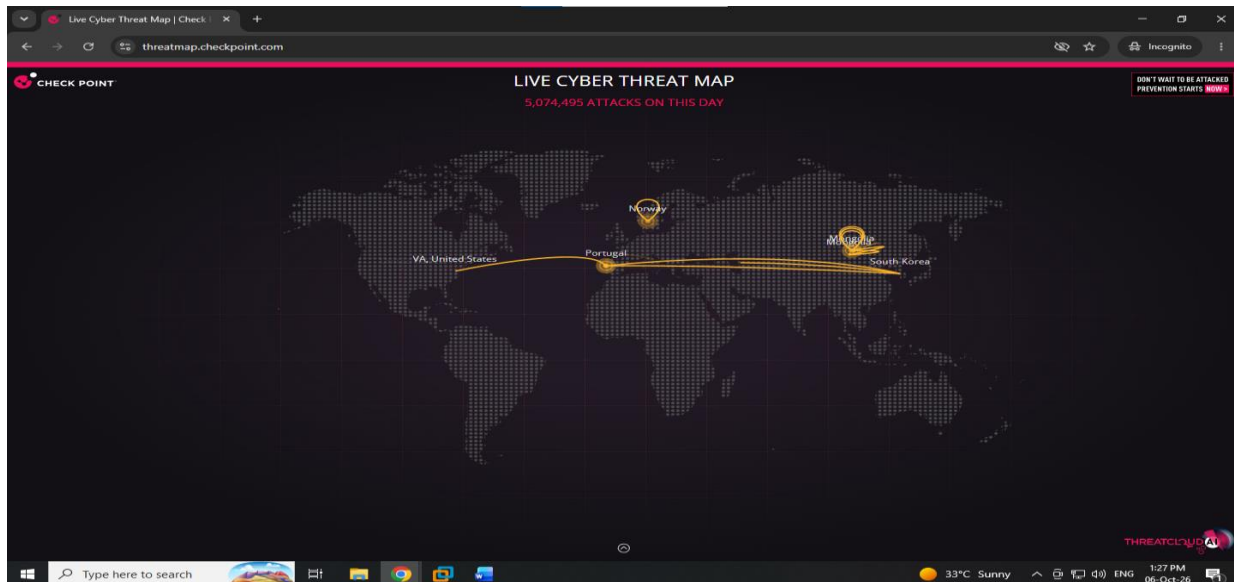
Q.3 Sign up for a free account on any online cybersecurity dashboard or threat intelligence tool (such as <https://www.cybermap.kaspersky.com/> or

ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

<https://threatmap.checkpoint.com/>). Take a screenshot of the dashboard and list 3 types of information or metrics you see displayed.

ANS :



ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

Q.4 Write a simple phishing email template targeting users of a popular app (for example, 'Your Paytm account is suspended, click here to verify'). Highlight the suspicious elements that make it a phishing attempt. Hint: Think about fake links, urgent language, or requests for sensitive info.

ANS :

From: security@quickpay-alerts.example

Subject: URGENT: Your QuickPay account has been suspended!

Dear Customer,

Your **QuickPay account has been temporarily suspended** due to unusual activity.

You must verify your account within 24 hours or your account may be permanently closed.

 **VERIFY YOUR ACCOUNT NOW**

To restore access, please provide:

- Your username and password
- Your debit/credit card details
- Your OTP

Thank you,
QuickPay Security Team

 **Suspicious elements**

- **Urgent language:** “URGENT” and “within 24 hours” pressure the recipient to act without thinking.
- **Fake-looking link:** The link uses a domain that is not the app's official website.
- **Request for sensitive information:** Legitimate services generally should not ask users to submit passwords or OTPs through an email link.
- **Threat of account closure:** The message tries to frighten the user into clicking.
- **Generic greeting:** “Dear Customer” instead of identifying the account holder.
- **Suspicious sender:** security@quickpay-alerts.example does not represent an official company domain.

ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

Q.5 Use ChatGPT or another AI tool to generate a simulated phishing email for a fictional food delivery app (like 'FoodiePal'). Copy the AI-generated email and explain which parts are likely to trick users and why.

ANS :

Urgent: Your FoodiePal account needs verification

From: FoodiePal Support support@foodiepal-security.example

Dear FoodiePal Customer,

We detected unusual activity on your account. For your security, your account has been temporarily restricted.

Please verify your account within 24 hours to avoid losing access to your FoodiePal account.

 **Verify My Account**

To complete the verification, you may be asked to confirm your account details.

If you did not request this verification, please contact FoodiePal Support immediately.

Thank you,

FoodiePal Security Team

Parts that could trick users

- **“Urgent” and “within 24 hours”** — creates pressure and makes users less likely to stop and verify the message.
- **“Unusual activity”** — sounds like a legitimate security warning and may make the recipient worry that their account has been hacked.
- **“Your account has been temporarily restricted”** — creates fear of losing access, encouraging the user to click.
- **“Verify My Account” button** — gives the user a convenient action instead of encouraging them to open the official app themselves.
- **Fake-looking website address** — foodiepal-security.example is not an official FoodiePal domain; this is a major warning sign.
- **Generic greeting (“Dear FoodiePal Customer”)** — legitimate account messages may contain identifying information, although a generic greeting alone does not prove phishing.

ASSIGNMENT : 1 – INTRODUCTION TO CYBERSECURITY

GUIDED BY : VIPUL MISTRI

- **Request to confirm account details** — phishing messages commonly attempt to collect credentials or other sensitive information.

Key lesson: If an email creates urgency and asks you to click a link to fix an account problem, don't use the link. Open the company's official app or website directly and check for notifications there.